DAILY TASK:--

1. Install nginx and run nginx on port number 81.

```
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

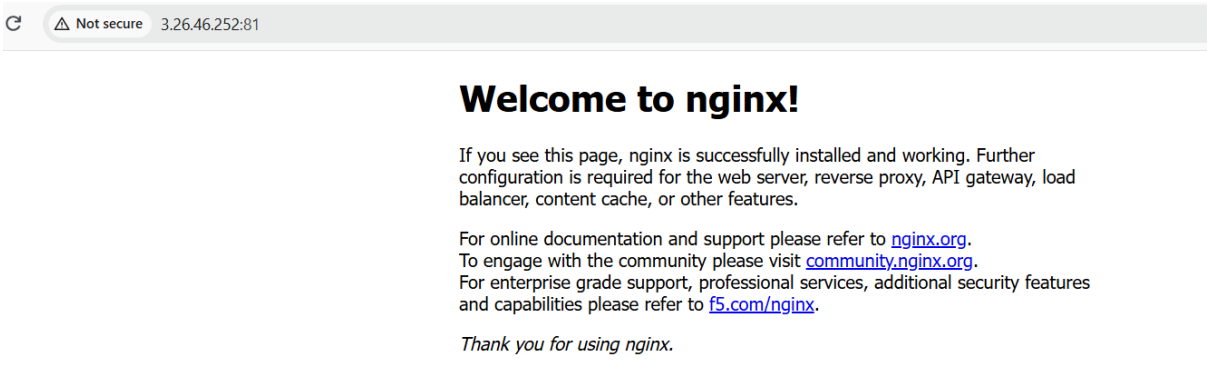
    server {
        listen       81;
        listen       [::]:81;
        server_name  _;
        root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

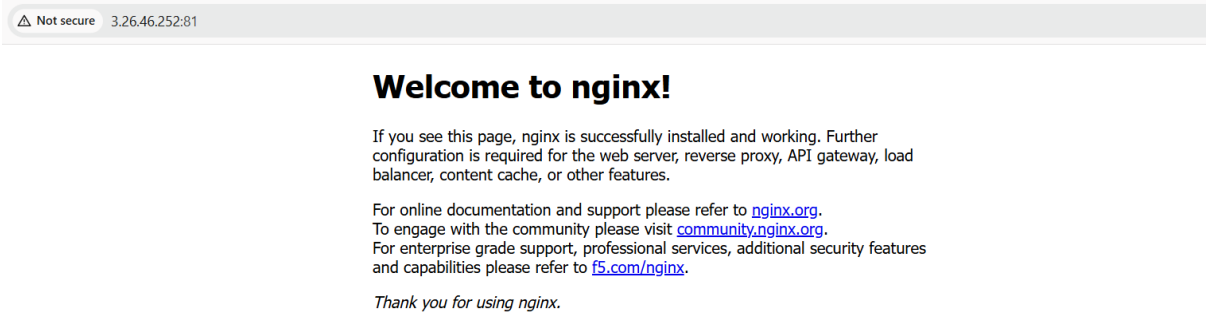
        error_page 404 /404.html;
        location = /404.html {
        }

        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }
    }

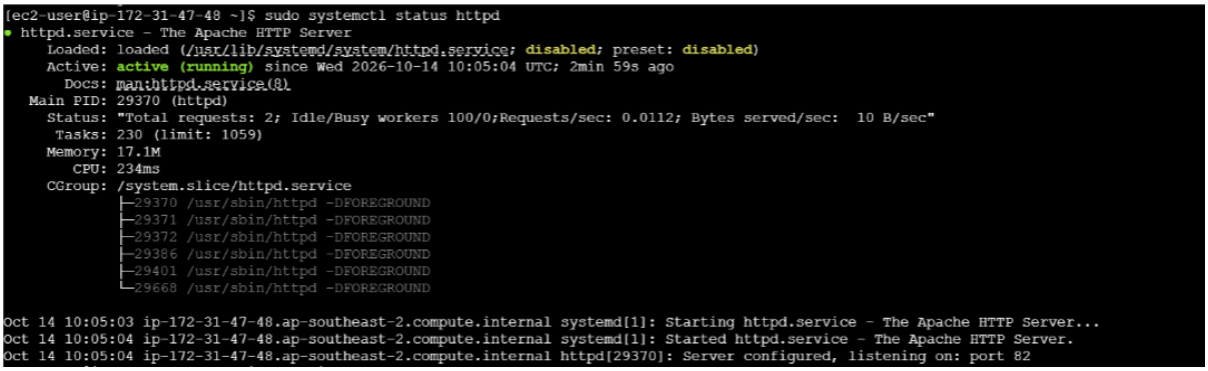
# Settings for a TLS enabled server.
#
#    server {
```



2. Deploy a sample index.html file on nginx.



3. Install Apache and run Apache on port number 82





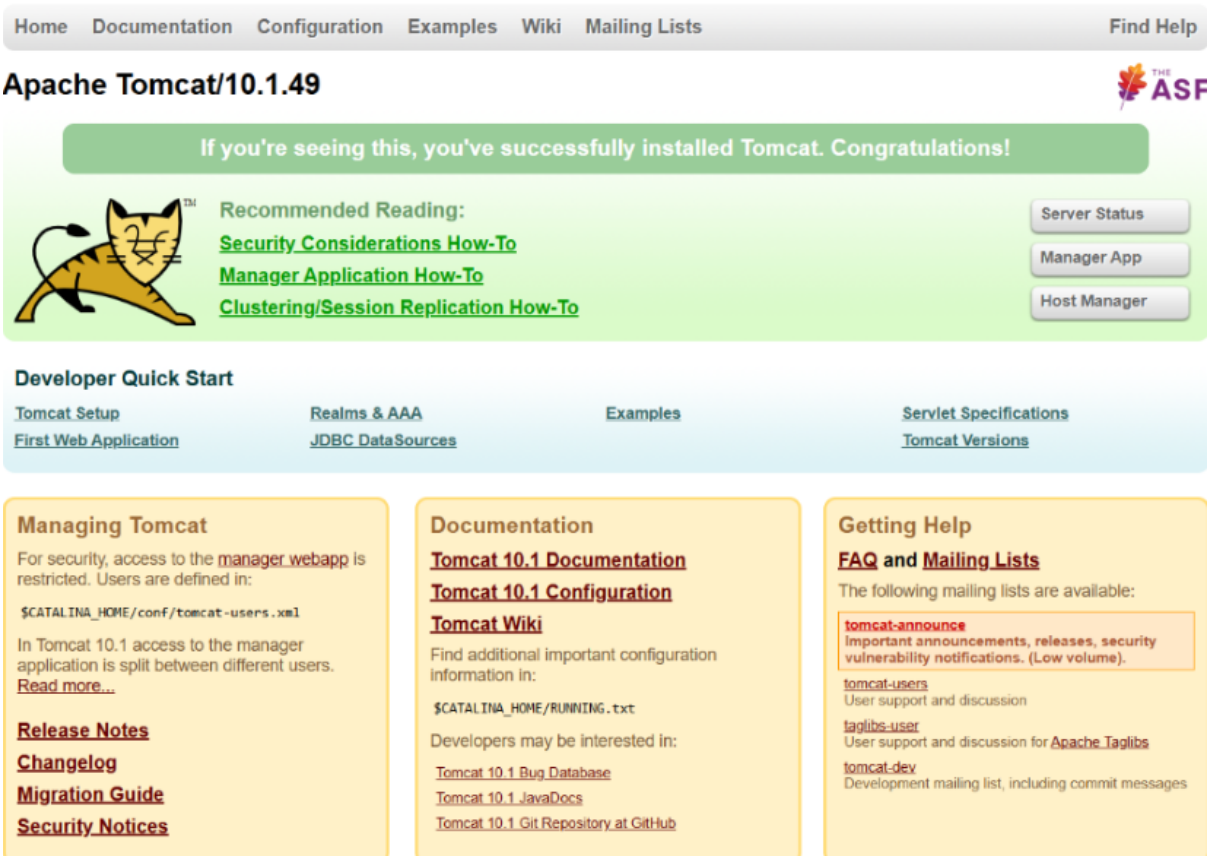
4.Deploy a sample index.html file on Apache.



It works!

5 . Install Apache tomcat on port number 8082

- Update Ubuntu system packages.
- Install Java JDK and verify the installation.
- Download Apache Tomcat.
- Extract Tomcat files into the /opt/tomcat directory.
- Open the Tomcat server.xml configuration file.
- Change the default port from 8080 to 8082.
- Save the configuration file.
- Start the Apache Tomcat server.
- Verify that Tomcat is running on port 8082.
- Allow port 8082 in the AWS EC2 Security Group.
- Open a web browser and access Tomcat using the EC2 public IP and port 8082.
- Verify that the Tomcat page is displayed successfully.



6. Deploy a sample app on webapps

```
    <p>Sample application deployed on Tomcat port 8082.</p>
</body>
</html>
ubuntu@ip-172-31-26-100:~$ sudo nano /opt/tomcat/webapps/sample/index.html
ubuntu@ip-172-31-26-100:~$ sudo cat /opt/tomcat/webapps/sample/index.html
<!DOCTYPE html>
<html>
<head>
    <title>Sample Tomcat Application</title>
</head>
<body>
    <h1>Welcome to Tomcat Server</h1>
    <p>Task 6 completed successfully.</p>
    <p>Sample application deployed on Tomcat port 8082.</p>
</body>
</html><!DOCTYPE html>
<html>
<head>
    <title>Sample Tomcat Application</title>
</head>
<body>
    <h1>Welcome to Tomcat Server</h1>
    <p>Task 6 completed successfully.</p>
    <p>Sample application deployed on Tomcat port 8082.</p>
</body>
</html>
ubuntu@ip-172-31-26-100:~$ sudo ss -lntp | grep 8082
LISTEN 0      100                *:8082            *:*    users:(("java",pid=10212,fd=43))
ubuntu@ip-172-31-26-100:~$ curl http://localhost:8082/sample/
<!DOCTYPE html>
<html>
<head>
```

Welcome to My Sample Application

This application is deployed on Apache Tomcat.

I navigated to the Tomcat webapps directory.
I created a new folder for my sample application.
I created a sample index.html file inside the application folder.
I added sample content to the application.
I restarted the Tomcat service.
I tested the sample application using port 8082.
I accessed the application using the EC2 public IP address and port 8082.
I verified that my sample application was displayed successfully.

7 .Create a tomcat.service file for tomcat.

```
  GNU nano 8.3                                          /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat 11 Web Application Server
After=network.target

[Service]
Type=simple

User=ec2-user
Group=ec2-user

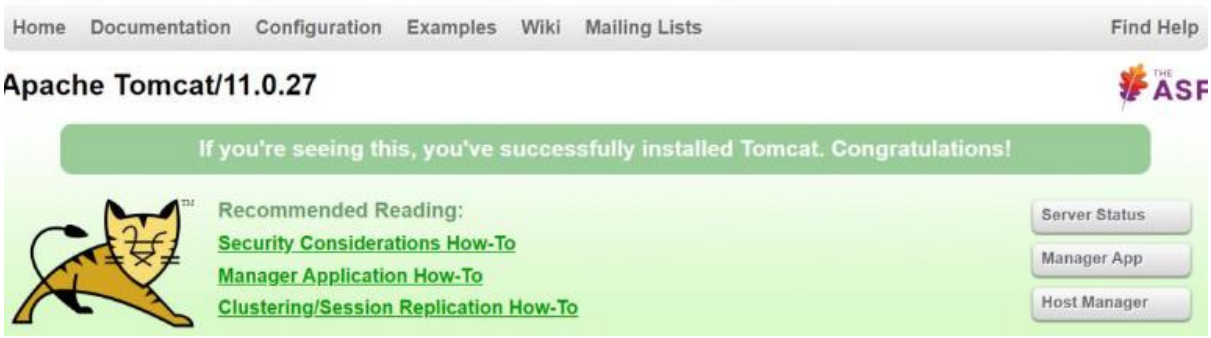
Environment="JAVA_HOME=/usr/lib/jvm/java-26-amazon-corretto.x86_64"
Environment="CATALINA_HOME=/home/ec2-user/tomcat/apache-tomcat-11.0.27"
Environment="CATALINA_BASE=/home/ec2-user/tomcat/apache-tomcat-11.0.27"

WorkingDirectory=/home/ec2-user/tomcat/apache-tomcat-11.0.27

ExecStart=/home/ec2-user/tomcat/apache-tomcat-11.0.27/bin/catalina.sh run
ExecStop=/bin/kill -s TERM $MAINPID

Restart=on-failure
RestartSec=5
TimeoutStopSec=30

[Install]
WantedBy=multi-user.target
```



8. Configure HA Proxy server

